The free locally convex space $L(\mathbb{N}^{\mathbb{N}})$ is not multi-Rosenthal

arXiv automatic math run `fa_banach_001`

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Source problem

The source paper is M. Komisarchik and M. Megrelishvili, *Tameness and Rosenthal type locally convex spaces*, arXiv:2203.02368. On page 33, Question 8.7 asks:

> Examine if $L(\mathbb{N}^{\mathbb{N}})$ is multi-Asplund or, at least, multi-Rosenthal.

Here $L(X)$ denotes the free locally convex space over the Tychonoff space X, and $\mathbb{N}^{\mathbb{N}}$ is the Baire space, homeomorphic to the space of irrationals P mentioned immediately before the question.

Definitions

A Banach space is called *Rosenthal* if it contains no isomorphic copy of ℓ_1. A locally convex space E is *multi-Rosenthal* if it is isomorphic, as a locally convex space, to a subspace of a product of Rosenthal Banach spaces. Similarly, E is *multi-Asplund* if it embeds into a product of Asplund Banach spaces.

Every Asplund Banach space is Rosenthal: if it contained ℓ_1, then the separable subspace ℓ_1 would have nonseparable dual ℓ_∞, contradicting the separable-subspace characterization of Asplund spaces. Therefore multi-Asplund implies multi-Rosenthal.

Idea of the Proof

The proof uses the universal property of $L(X)$ in the most direct possible way. If $L(\mathbb{N}^{\mathbb{N}})$ were multi-Rosenthal, then every Banach-valued continuous map on $\mathbb{N}^{\mathbb{N}}$ would extend to a continuous linear operator on $L(\mathbb{N}^{\mathbb{N}})$, and that operator would be controlled by finitely many coordinates of the product of Rosenthal spaces. Thus the operator would factor through a Rosenthal Banach space.

But the Baire space maps continuously onto every Polish space, in particular onto ℓ_1. Extending such a surjection would force ℓ_1 to be a quotient of a Rosenthal Banach space. This is impossible, because the property of containing no ℓ_1 passes to Banach quotients.

Auxiliary facts

Lemma 1 (finite-coordinate factorization)**.** *Let E be a multi-Rosenthal locally convex space and let $T : E \to B$ be a continuous linear map into a Banach space. Then T factors through a Rosenthal Banach space.*

Proof. Embed E as a locally convex subspace of a product $\prod_{i \in I} R_i$, where each R_i is a Rosenthal Banach space. Since T is continuous into a normed space, there are finitely many coordinates $F \subset I$ and a constant C such that

$$\|Tx\| \leq C \max_{i \in F} \|x_i\|_{R_i}$$

for every $x \in E$. Hence T vanishes on the kernel of the coordinate projection $\pi_F : E \to \prod_{i \in F} R_i$, and so $T = S\pi_F$ for a bounded linear map S on $\pi_F(E)$. Extending S to the closure

$$G = \overline{\pi_F(E)} \subseteq \prod_{i \in F} R_i$$

gives a factorization $E \to G \to B$.

The finite product of Rosenthal Banach spaces is Rosenthal, and closed subspaces of Rosenthal Banach spaces are Rosenthal. Thus G is Rosenthal. □

Lemma 2 (quotients)**.** *A Banach quotient of a Rosenthal Banach space is Rosenthal.*

Proof. Let $q : X \to Y$ be a quotient map, and suppose that Y contains an isomorphic copy $J : \ell_1 \to Y$. By the open mapping theorem, there is $C > 0$ such that for every $y \in Y$ one can choose $x \in X$ with $qx = y$ and $\|x\| \leq C\|y\|$. Choose $x_n \in X$ with $qx_n = Je_n$ and $\sup_n \|x_n\| < \infty$.

Define $S : \ell_1 \to X$ by

$$S(a) = \sum_{n=1}^{\infty} a_n x_n.$$

The series converges absolutely, so S is bounded, and $qS = J$. Since J is an isomorphic embedding,

$$\|Sa\| \geq \frac{1}{\|q\|} \|Ja\|$$

for every $a \in \ell_1$. Hence S embeds ℓ_1 into X. Thus if X contains no copy of ℓ_1, neither can Y. □

Main theorem

Theorem 1. *The free locally convex space $L(\mathbb{N}^{\mathbb{N}})$ is not multi-Rosenthal. Consequently it is not multi-Asplund.*

Proof. The Baire space $\mathbb{N}^{\mathbb{N}}$ admits a continuous surjection

$$s : \mathbb{N}^{\mathbb{N}} \to \ell_1,$$

because every Polish space is a continuous image of $\mathbb{N}^{\mathbb{N}}$.

By the universal property of the free locally convex space, s extends to a continuous linear operator

$$\widehat{s} : L(\mathbb{N}^{\mathbb{N}}) \to \ell_1$$

with $\widehat{s}(\delta_x) = s(x)$. Since s is onto, $\widehat{s}$ is onto.

Assume, toward a contradiction, that $L(\mathbb{N}^{\mathbb{N}})$ is multi-Rosenthal. By Lemma 1, $\widehat{s}$ factors as

$$L(\mathbb{N}^{\mathbb{N}}) \longrightarrow G \xrightarrow{u} \ell_1$$

where G is a Rosenthal Banach space and u is bounded linear. Since $\widehat{s}$ is onto, $u(G) = \ell_1$. Therefore ℓ_1 is a Banach quotient of the Rosenthal Banach space G, contradicting Lemma 2. Thus $L(\mathbb{N}^{\mathbb{N}})$ is not multi-Rosenthal.

Finally, every Asplund Banach space is Rosenthal, so a multi-Asplund embedding would in particular be a multi-Rosenthal embedding. Hence $L(\mathbb{N}^{\mathbb{N}})$ is not multi-Asplund. □

Verification notes

The proof uses only standard permanence properties and one classical descriptive-set-theoretic fact: every Polish space is a continuous image of the Baire space. The quotient permanence of Rosenthal Banach spaces is proved above using the projectivity of ℓ_1, unfolded into an explicit lifting of its unit vector basis.

The result is stronger than the previously known statement quoted by Komisarchik–Megrelishvili that $L(P)$, with P the irrationals, is not multi-reflexive. It rules out embeddings into products of all Rosenthal Banach spaces, not only reflexive Banach spaces.

Novelty and literature check

Local run indexes were searched for arXiv:2203.02368, the source title, Question 8.7, $L(\mathbb{N}^{\mathbb{N}})$, multi-Asplund, multi-Rosenthal, and free locally convex space. No existing packet or attempt for this question was found.

Bounded web searches on 2026-06-15 for exact phrases including "$L(\mathbb{N}^{\mathbb{N}})$" with multi-Rosenthal, multi-Asplund with free locally convex spaces, and the source title with multi-Rosenthal found the source paper and nearby multi-reflexive/nuclear literature, but no prior answer to Question 8.7. The novelty confidence is moderate: the argument is short and may be known to specialists, but no exact published answer was located in this search.

References

- M. Komisarchik and M. Megrelishvili, *Tameness and Rosenthal type locally convex spaces*, arXiv:2203.02368.
- A. Leiderman and V. Uspenskij, *Is the free locally convex space $L(X)$ nuclear?*, arXiv:2106.13413.
- J. Diestel, *Sequences and Series in Banach Spaces*, for standard Rosenthal-space and ℓ_1 background.
- A. S. Kechris, *Classical Descriptive Set Theory*, for the classical fact that every Polish space is a continuous image of the Baire space.

Human review recommendation

This packet should be reviewed as a likely full negative answer to Question 8.7. The main points to check are Lemma 1, especially the finite-coordinate domination step for maps from a subspace of a product, and Lemma 2, the quotient permanence of the Rosenthal property.